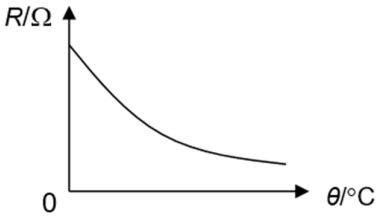
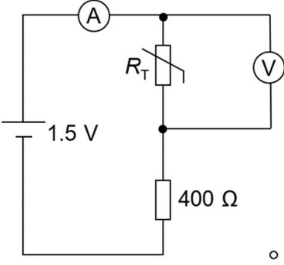


Qn	Solution
1(a)(i)	The potential difference between two points in a circuit is the <u>energy converted from electrical to other forms per unit charge</u> passing from one point to the other.
1(a)(ii)	Electromotive force of a source is <u>the energy converted from other forms to electrical per unit charge</u> driven through the source
1(b)	
1(c)(i)	
1(c)(ii)	$V = \frac{400}{400 + 30}(1.5) = 1.3953 \approx 1.4 \text{ V}$
1(d)(i)	$V_T = 1.5 - 1.3953 = 0.10465 \approx 0.10 \text{ V}$ $P_T = \frac{V_T^2}{R_T} = \frac{0.10465^2}{30} = 3.6506 \times 10^{-4} \approx 3.7 \times 10^{-4} \text{ W}$
1(d)(ii)	$V = \frac{1.2}{1.2 + 30}(1.5) = 0.057692 \approx 0.058 \text{ V}$ $V_T = 1.5 - 0.057692 = 1.4423 \approx 1.4 \text{ V}$ $P_T = \frac{V_T^2}{R_T} = \frac{1.4423^2}{30} = 0.069341 \approx 0.069 \text{ W}$ p.d. across the resistor will drop to 0.058 V; p.d. across the thermistor will increase from 0.1 V to 1.4 V. Thermal power dissipated at the thermistor will increase to 0.069 W. This cause an significant increase in temperature which will cause the resistance of thermistor to decrease; and p.d across the thermistor will then start to decrease.

$2(a)$	Γ	h
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